ARCUS

INFRASTRUCTURE INTELLIGENCE

Territory Briefing: Roma

Professional province-level screening based on ARCUS historical evidence, territorial exposure, ARCUS/AINOP Collapse Rate and historical-pattern reading.

PRIORITY

52/100

COLLAPSE

0.99x / national rank 41

EVENTS

5

DECISION MESSAGE

Distributed chronic risk

REPORT DATE

Jun 10, 2026

CONTEXT

Bridge

PROVINCE

Roma

OUTPUT

New construction

ARCUS is built on the first systematic database of collapsed bridges in Italy from 2000 to today: the report does not merely rank cases, it reads documented historical patterns to guide preliminary checks.

01 EXECUTIVE SUMMARY

The selected province requires preliminary screening for Bridge interventions. The value is not the case list: it is understanding whether Roma has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

RECOMMENDED USE

ARCUS reading: for new Bridge interventions, the Impact signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Roma.

PRIORITY INDEX

52/100

exposure index + heritage reading

COLLAPSE RATE

0.99x

ARCUS/AINOP national benchmark; confidence high

HISTORICAL EVENTS

5

Impact dominant

RELIABILITY

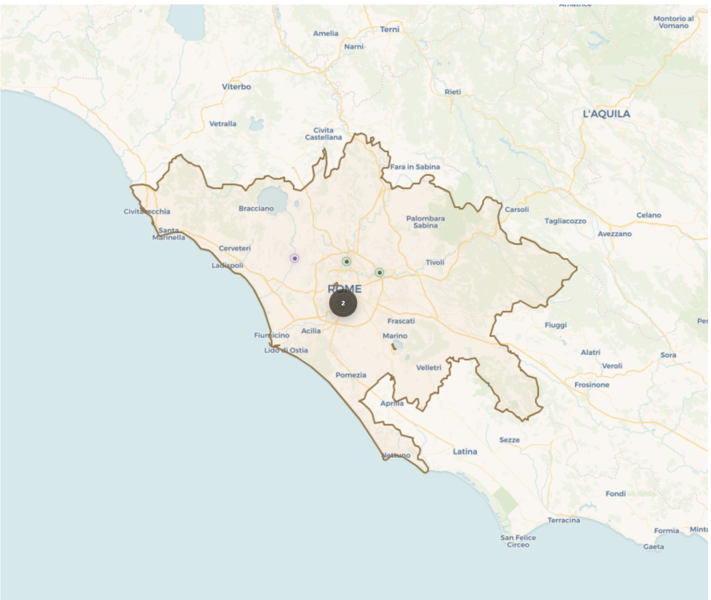
84/100

14 linked sources

COLLAPSE RATE INTERPRETATION

The 0.99x rate versus the national average places Roma among the provinces with documented historical vulnerability above the ARCUS/AINOP benchmark. Roma is national rank 41 by documented ARCUS/AINOP collapse rate. Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

02 TERRITORY READING & MAP



ROMA / ARCUS ATLAS EXTRACT

GEOMORPHOLOGY AND CONTEXT

Roma shows a heterogeneous territorial profile: 5 ARCUS events across 2 municipalities indicate that the structural driver must be read together with local morphology, drainage network, slopes and crossing continuity.

MAP READING

The map locates 5 ARCUS cases across 2 municipalities and should be read with the historical pattern: Distributed chronic risk. Dominant date: 2014-07-24.

03 HISTORICAL FAILURE CONTEXT

HISTORICAL PATTERN

Distributed chronic risk. historical risk distributed across multiple events and dates (20% on the most recurrent date).

DOMINANT TRIGGER

60%

Impact

HAZARD

structural

dominant territorial signal

READING

distributed risk

20% on the most recurrent date; read as distributed temporal risk plus local event clusters.

04 ACTIONS

- Use the territorial exposure indicators to identify the dominant territorial signal.
- Read the documented distributed historical pattern and use individual cases as map/appendix references.
- Request the minimum technical data package for the dominant structural signal: local hazard study, asset geometry, foundation/abutment exposure, inspection history and source-backed site constraints.
- Commission site-specific checks calibrated to the dominant structural context and local geomorphology before moving from provincial screening to design assumptions.
- Use CSV and GeoJSON exports for technical coordination.

DATA REQUEST PACKAGE

Local hazard study; asset geometry and foundations; inspection history; source-backed site constraints.

COLLAPSE RATE CONFIDENCE NOTE

Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

05 EVENT REFERENCE APPENDIX

The following list is a documentary reference, not an operational P1/P2/P3 ranking. The decision message remains the provincial historical pattern.

ID	Date	Municipality	Severity	Cause
B14.07.02	Jul 24, 2014	Roma	Total Collapse	Impact
B20.09.01	Sep 25, 2020	Santa Maria di Galeria	Total Collapse	Overload
B20.10.12	Oct 03, 2020	Roma	Total Collapse	Impact
B21.10.01	Oct 02, 2021	Roma	Total Collapse	Fire and Explosion
B13.09.01	Sep 26, 2013	Roma	Total Collapse	Impact

06 METHODOLOGY SNAPSHOT

Layer	Meaning
Priority Index	Territorial exposure index: synthesizes dominant hazard, ARCUS case density, event concentration and evidence reliability. It does not directly measure historical bridge-stock vulnerability.
Collapse Rate	Ratio between ARCUS cases and AINOP counted bridges in the province. Value: 0.99x, national rank 41; it should be read separately from the Priority Index.
Why Separate	Priority Index indicates where the territory has elevated hazard exposure; Collapse Rate indicates where the bridge stock has shown historical vulnerability above or below the national average.
Historical Failure Context	Historical-pattern reading: concentrated event or distributed risk over time.
Hydraulic Cause Category	Hydraulic cause category: ARCUS uses 'hydraulic' as the documented trigger when sources confirm a flood or high-water event as the proximate cause. Mechanism-level classification - scour, bank erosion, debris impact, overtopping - is applied only where primary technical sources provide
Evidence Window	Observed period: ARCUS database of collapsed bridges in Italy from 2000 to today; close events in the same time window are read as one scenario, not decision duplicates.
AINOP Coverage	Collapse Rate uses AINOP counted bridges as provincial denominator. Confidence: high; local undercounting may inflate the ratio.
Evidence Classes	A = primary institutional/technical source; B = reliable technical or media source; C = generic documentary evidence; D = weak evidence to strengthen.